

On a Study of Layout Aesthetics for Business Process Models Using BPMN

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Abstract. As BPMN spreads among a constantly growing user group, it is indispensable to analyze the expectations of users towards the appearance of a BPMN model. The user groups are mostly inhomogeneous since users stem from different backgrounds, e.g. IT, managerial sciences or economics. It is conceivable that BPMN novices may have different issues compared to higher skilled modeling experts. When this large set of users starts modeling, the expectations considering the graphical outcome of the modeling process may differ significantly.

In this work, we analyze layout preferences of user groups when modeling with BPMN. We present a set of layout criteria that are formalized and then confirmed by a user study. The conduction of the study reveals preferences of single user groups with respect to secondary notation and layout aesthetics. From our results, proposals for adaptations of software tools towards different BPMN users can be derived.

Keywords: user study, secondary notation, BPMN models, layout aesthetics.

1 Motivation

BPMN is a visual modeling language for business processes. The modeling toolbox comprehends elements such as activities, gateways, events etc. and connecting elements, e.g. associations, sequence or message flows. Also, BPMN provides pools and swimlanes which represent structural elements. When users begin to design a model they start on a blank sheet, or screen respectively, which implies that there are no hints for the user where to position the elements.

The development of an automatic layout algorithm that supports the user in this task is a mere logical progression. The layout algorithm could point to possible positions of new elements or could recalculate a 'good' layout from a given sketch. In [1,2], approaches for BPMN layout algorithms are undertaken. However, when developing a layout algorithm for a specific language as BPMN, one finds himself quickly in the state of asking: What actually matters a layout to be 'good'? What are the attributes that render a layout of a business process model to be readable and easy to grasp? What preferences or expectations towards a layout for business process models does a user have?

From this starting point, we designed an empirical user study with the aim to reveal the attributes that actually matter for BPMN users when modeling and looking for ‘good’ layout of their diagrams. In this work, we refer to those attributes as layout aesthetics, as proposed in [3,4]. Our study also takes into account that modeling users of BPMN are an inhomogeneous set of people with diverse background and skills, e.g. education in process modeling or modeling experience in practice are important measurement factors [5].

From a cognitive point of view, layout aesthetics are known as *secondary notation* [5,6]. In contrast to *first notation*, in our case the graphical notation of BPMN, secondary notation carries extra information by other means than the official syntax [7]. According to [7], ‘redundant recoding gives a separate and easier channel for information that is already present in the official syntax. Escape from formalism allows extra information to be added, not present in the official syntax’. As extra information in secondary notation, layout aesthetics are non-obvious cues in a graphical representation of a business process model. Although single aesthetics are not visible in a diagram but rather hidden in the layout, they are highly relevant for exploratory and modification activities [7] which both are very frequent tasks in the field of business process modeling.

In the user study, we support secondary notation in terms of layout aesthetics for BPMN. Moreover, we show preferences of specific user groups of modeling participants. We analyze layout aesthetics of users in comparison groups filtered by attributes ‘gender’, ‘experience’ and ‘education’. The results can be used for designing user-supporting features in modeling tools and tools that contain layout algorithms and produce well-received layout for diagrams. The rest of the paper is organized as follows: In Section 2, we present our formal catalogue of layout aesthetics, followed by the summary of our test method in Section 3 and the presentation of our study conjectures and their evaluation in Section 4. Before we conclude, we give an overview on related research work in Section 5.

2 Layout Aesthetics

Layout aesthetics are a measure for a graphical property of a drawing. The goal of a layout algorithm is to fulfill given aesthetics or, in other words, to optimize the aesthetics requirements. For the development of an algorithm, layout aesthetics must be formalized and modeled in the algorithm. The formalizations must be known beforehand in order to adapt the layout algorithm accordingly.

For BPMN, results of research considering specific aesthetics are not known yet. However, for diagrams that are also graph-based, aesthetics were explored in studies, e.g. for diagrams in general [8] or specifically for UML-diagrams [9]. Since we consider BPMN to be a graph-based notation, the underlying structures correspond to graphs with BPMN elements as nodes and BPMN connection elements as edges. For general graphs, we can find aesthetics in the graph-drawing community that discusses general aesthetics. However, general graphs don’t possess notation semantics, in contrast to BPMN where the graphical representation defines the type of the BPMN element, e.g. gateways have diamond shapes.

In [10], an overview on common graph drawing aesthetic criteria is given. Also, in [5], a subset of these criteria are confirmed to be important for BPMN.

The formalization of layout aesthetics is reached by expressing them as optimization problems, as in [3,4,11]. For BPMN as a graph-based notation, we consider the following layout aesthetics as necessary:

- Minimize the number of crossings of connecting elements (CROSSING).
- Minimize the area of the drawing (AREA).
- Minimize the number of bends of connecting elements (BEND).
- Minimize the number of overlapping (connection) elements (OVERLAP).
- Maximize the number of orthogonally drawn connecting objects (ORTHOGONAL).
- Maximize the number of connecting objects respecting workflow direction (FLOW).

Therefore, layout algorithms solve a multi-objective optimization problem. However, the task to tackle the whole set in a single algorithm is very complex which is underlined by the low number of tools supporting BPMN layout and the poor quality of layout results, see [1,12] for a quick overview.

Also, BPMN has specific requirements towards layout aesthetics since it provides notation semantics within its graphical representations. The following requirements represent aesthetics that consider the specific requirements of BPMN which can be derived inspecting the standardization document:

- Flow objects have different sizes (ELEMENT_SIZE).
- Partitions must be considered, e.g. pools and swimlanes (PARTITION).
- Labeling of pools, swimlanes and (connection) elements must be feasible (LABEL).

These principles are also mentioned in informal collections of drawing principles for special diagram types, e.g. for UML activity diagrams [13] or network diagrams [14]. Summing up the relevant aesthetics, we state the following list of standard layout aesthetics for the layout of BPMN diagrams:

CROSSING, ORTHOGONAL, AREA, BEND, PARTITION, LABEL, ELEMENT_SIZE, OVERLAP and FLOW.

The list of design standards is also conform on conventions within the BPMN community [1,15] and corresponds to a superset of supported aesthetics in existing BPMN tools.

In our test design, we developed a catalogue with statements that cover the different objectives of layout aesthetics. We face this catalogue with the users' personal appreciation of different layouts in order to confirm or reject our conjectures which will be proposed in Section 4.1.

3 Design of the User Study

3.1 Software Tools

In this Section, we will give a summary of our test design and test method. For a more detailed description, we refer to [12]. For the comparison of layout results for BPMN models, we have to choose among existing solutions of tools that support modeling with BPMN and provide automatic layout features. After a market study, we obtained a list with 54 software packages that support BPMN according to the vendors. However, support of BPMN only is not sufficient for our study, therefore, we developed the following criteria for tool selection:

1. BPMN support:
The tool must support the standardized version 1.2 of BPMN, including all elements given in the standard.
2. Automatic layout support:
The tool must provide the user with a feature that calculates an automatic layout of a given BPMN model. Thus, the user has the possibility to acquaint himself with the principles of layout aesthetics. The layout feature must also consider basic BPMN aesthetics, e.g. swimlanes must be respected and edges must be drawn orthogonally.
3. Evaluation license availability:
For our study, we depend on evaluation licenses provided by the software vendors.

These criteria were compulsory. We could not consider tools or vendors that could not meet one or more of the criteria. During the study preparation, the most challenging criteria was the support of automatic layout. Also, software that supposedly supports BPMN provided repeatedly only a subset of BPMN. This prevented us from preparing our modeling test case on those tools.

After applying our criteria filters on the list of software tools, we obtained a set of five tools (three commercial tools plus two versions of a self-developed tool [1]) that fulfilled all criteria and that were considered for our user study. In [12], we provide a complete list with all software vendors that were considered.

3.2 Test Method

In the following, we will shortly describe the procedure instructions given to probands and the surrounding setup for the experiment of the study. The setup was successfully confirmed to be well-chosen with the help of a pre-test prior to the experiment.

The probands of the study were chosen among students with majors in economics and/or computer science. Their skills (education and/or experience) in process modeling spread from very low to very high in order to represent inhomogeneous but, because of their major subject, potential future users of modeling tools.

The software tools were installed on PCs and probands were randomly assigned to one tool. For assuring a basic common understanding on BPMN and business process modeling, probands initially were asked to insert a simple extension into an existing process using the assigned modeling tool. The set of BPMN elements needed for the implementation of the extension was given as support. The process for the modeling task was a book order instance. The process can be inspected in detail in Figure 6. After the modeling part, probands were requested to prepare a presentable version of the new process. They were advised to use the automatic layout feature(s) of the corresponding tool. At this point, probands were supposed to be familiar with BPMN modeling and also with the automatic layout results and features of the respective tool. For the evaluation, each proband obtained five different layout versions of the new process, one from each evaluated tool. In the accompanying questionnaire, probands were asked to give rankings among the tools for each statement contained in the questionnaire. For comparison reasons, we chose to ask for rankings with strictly relative order among the tools. Thus, two tools cannot be ranked equally and we obtain more exact responses. The statements of the questionnaire correspond to the layout aesthetics of Section 2. All layout aesthetics are represented by a statement. For the representation as statements, we group the statements into categories that aimed at the same attribute of a graph/diagram. The following list gives the wording of the statements as they appeared in the questionnaire:

1. Category 1: “Connection elements” (edges)
 - 1.1 Edges are drawn orthogonally and are inserted in such a way that they appear as short as possible (ORTHOGONAL).
 - 1.2 Edges appear to be drawn with the lowest possible number of crossings (CROSSING).
 - 1.3 Edges appear to be drawn with the lowest possible number of bends (BEND).
 - 1.4 Edges are drawn such that they consider the reading direction (FLOW).
2. Category 2: “Area usage”
 - 2.1 The size of the swimlanes is chosen such that all elements have enough space (ELEMENT_SIZE).
 - 2.2 The diagram contains unused space that could be better exploited by rearranging the elements (AREA).
3. Category 3: “Elements”
 - 3.1 Elements are arranged such that they do not overlap (OVERLAP).
 - 3.2 The size of the elements is chosen such that the description of the label is readable (LABEL).
 - 3.3 The assignment of an element to its swimlane is easy to perceive (PARTITION).
4. Category 4: “Coloring”
 - 4.1 The choice of colors supports to obtain a detailed comprehension of the model.
 - 4.2 The choice of colors supports to obtain a quick overview of the model.

After the ranking of tools for each statement, probands were asked to rank the categories in decreasing importance according to their personal judgement. Also, probands were encouraged to add additional categories and/or statements that to their opinion were not represented in our above catalogue of statements. These proband-defined categories were also included in the ranking of categories. Eventually, the probands are asked to give a ranking of all layout diagrams presented in the file to their personal appearance preference, independently from the above given statements.

The total time per proband of our study depended on the skill level of the probands. The periods of time ranged from 55-75 minutes per proband for the task of modeling and responses to the two questionnaires that considered layout and usability questions. The total number of participants of the study was 39.

4 Evaluation

In this section, we present the results of our study. The study data was collected while conducting the study at two universities, at Eberhard Karls Universität Tübingen and Humboldt Universität Berlin. The circumstances were kept comparable during both times, identical software tools and identical questionnaires were used.

Note that generally on ordinal scales, the application of mean values is not possible. However, in our case, as stated in [16], the distance between the ordinal items can be considered to be equal from a proband's perspective. Therefore, the application of mean values for our evaluation is rendered possible and allows an thorough evaluation using descriptive statistics.

First, we analyze the categories of Section 3.2. For the categories, we present the ranking of all users. For the analysis, we also examine the results of the proband group separately by gender, process modeling experience and prior education in modeling.

For the analysis, the total user group is filtered yielding to subset groups with the following attributes:

- Gender: male (m) and female (f)
- Experience: business process modeling experience is rated from 'none' (0) and 'low' (1) to 'high/very high' (2+).
- Education: number of lectures (or similar events) attended in the field of business process modeling; the group is split into ranges from none (0), one (1) to two or more (2+).

The division into subset groups is due to the focus on inhomogeneous user groups. Each subset group represents a set of business process modelers with distinct backgrounds, capabilities and skills. Thus, we analyze the corresponding preferences of these groups. Before that, we propose our conjectures for the analysis in the following section.

4.1 Conjectures

Our conjectures aim to confirm that the set of statements that we defined in Section 3.2 is convenient to support the aesthetics from Section 2 and correspond to the users' preferences.

For better coverage, we state our conjectures for the total group and the subset groups separately. The conjectures will also be evaluated separately in Section 4.3.

Conjecture 1 (Layout Aesthetics for Total Group). The user rating of the total group with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) correspond.

If Conjecture 1 is supported, we can state that the statements of our catalogue of categories 1-4 are sufficient to fulfill the aesthetics' requirements of a BPMN diagram for all probands.

Conjecture 2 (Layout Aesthetics for Subset Group 'Gender'). The user rating of the subset groups 'Gender' (male/female) with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) correspond.

If Conjecture 2 is supported, we can confirm that the statements of our catalogue of categories 1-4 are sufficient to fulfill the aesthetics' requirements for the subset group of male, or female respectively, probands.

Conjecture 3 (Layout Aesthetics for Subset Group 'Experience'). The user rating of the subset groups 'Experience' (Experience (0), Experience (1), Experience (2+)) with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) correspond.

Conjecture 3 states the validity of the statements considering probands with different levels of modeling experience, from none (novices) to very high (experts).

Conjecture 4 (Layout Aesthetics for Subset Group 'Education'). The user rating of the subset groups 'Education' (Education (0), Education (1), Education (2+)) with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) correspond.

If Conjecture 4 is supported, we can confirm that the statements are sufficient to fulfill the aesthetics' requirements for the subset groups of probands with different educational background for business process modeling.

If the correspondence between the general ranking and the catalogue of statements is not given, that means that Conjectures 1- 4 cannot be supported, we state a conjecture for the case that, from the resulting values, we can derive a tendency to the better- and the less well-rated tools.

Conjecture 5 (Layout Aesthetics and Tool Tendency). From the user ratings of the Total Group/Subset groups 'Experience'/'Education'/'Gender', we can observe a match between the best- and the worst-rated tools with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking).

In addition to conjectures on whole subset groups, we state conjectures that claim the statements of our layout aesthetics adapt better the higher the users' experience or the better their education in business process modeling with BPMN. In other words, the differences between the general ranking and the catalogue of aesthetics' statements diminish inside the subset group 'Education', or 'Experience' respectively:

Conjecture 6 (Increasing Benefit with Higher Experience). The differences in the user rating of the subset groups 'Experience' with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) diminish with increasing experience levels of probands.

Conjecture 7 (Increasing Benefit with Higher Education). The differences in the user rating of the subset groups 'Education' with respect to 1) the catalogue of category statements and 2) the users' personal appreciation (general ranking) diminish with increasing educational background of probands.

Conjectures 6 and 7 represent our conjecture that our layout aesthetics are more efficient and exceedingly appropriate for BPMN modeling users that have a fundamental understanding of process modeling in general. Before analyzing our set of conjectures, we present the results of the ranking of categories.

4.2 Results of Category Ranking

The results of the category ranking show where the subset groups see the aesthetic with the largest impact on the layout. The categories were ranked in a relative order from 1 to 4, respectively to 5 or 6 if proband-defined categories were given. The diagram in Figure 1 depicts the resulting values. After normalization, values can range from 0 to 4 while 4 is the highest (best) score. The following interpretation of Figure 1 is done for each category.

For category 1 "Connection elements", the values contain only two spikes, for group 'Experience (1)' and group 'Experience (2+)'. However, the derivations of the spikes are diametrically opposed. Thus, no conclusion can be drawn from these spikes. However, together with the highest mean value of all categories, the low number of derivations show that category 1 has a great effect on the acceptance of the users. Thus, we can state that users prefer layout models that correspond with statements 1.1 - 1.4.

Category 2 "Area usage" has the largest spikes in the 'Education' subset group, more precisely it has an obvious peak in 'Education (0)'. This peak renders category 2 to be the most important for 'Education (0)', in contrast to other subset groups. For subset groups 'Education (1)' and 'Education (2+)',

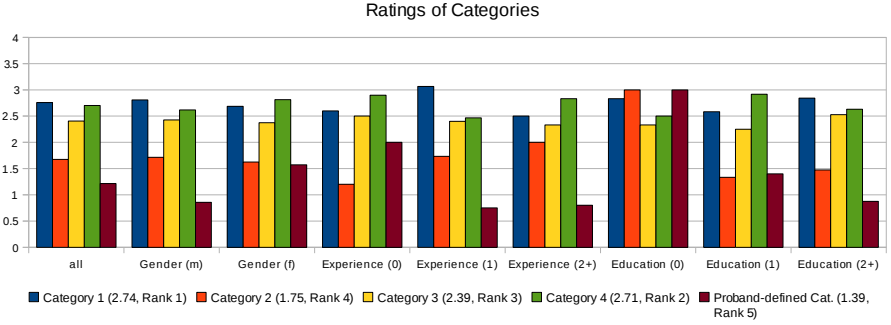


Fig. 1. Ratings of the categories for total group (all) and all subset groups. Mean values and ranks for categories are given in parentheses.

the derivations are low. Thus, category 2 seems to lead to a most promising but eventually alleged effect on the layout for users with less education in process modeling. In total, category 2 ranks 4th and therefore last of all pre-defined categories.

Category 3 “Elements” is the category with the significantly least derivations in all subset groups. Thus, all users agree that element layout properties are indispensable, but no group rated this category 1st, in other words most preferred. Rank 3 for this category shows that it contributes important statements and may not be ignored when creating layout models.

Category 4 “Coloring” has only two minor spikes. In subset groups ‘Experience (1)’ and ‘Education (1)’, the spikes show two derivations to slightly higher, or lower respectively, values. However, in most subset groups, category 4 is rated most important or second most important after category 1. Thus, together with category 1, this category has strong effect on the layout.

The proband-defined category shows the most diverging behaviour. While for subset group ‘Education (0)’, the category yields the highest value, it ranks last for 6 of 9 subset groups. Only subset group ‘Experience (0)’ describes a similar behaviour as ‘Education (0)’. The value of ‘Experience’ and ‘Education’ may already imply that the more experience or education modeling users are equipped with, the more the rating of models converges to a set of commonly accepted measurement attributes and aesthetic criteria (Conjectures 6 and 7).

4.3 Overall Ranking

We will now face the results of the general ranking given by the probands with the aggregation over the results of the set of categories. Therefore, we calculate the mean values of the general ranking for both, the total group and each subset groups. This approach allows to consider the subset groups and their inhomogeneities. For obtaining the results, we compare data from the general ranking (GR) over all tool models with the aggregated data per category and its corresponding statements from Section 3.2 (mean of categories, MC1-4).

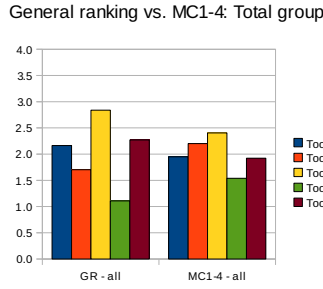


Fig. 2. Comparison of general ranking (GR) and mean of categories (MC1-4) for total group

By Total Group:

The contrast between the GR and MC1-4 is depicted in Figure 2 and becomes visible when taking a closer look at Tools A,B and E. Their ranking is not equal over both measurements. However, as Tool C always ranks 1st and Tool D ranks last, we can state that our methods can distinct between tools that suffice users' preferences and those that fail. Thus, we have to reject Conjecture 1, but the support of Conjecture 5 for the total group is successful.

By Gender:

The gender subset group contains the male and female subsets of probands. For the comparison between GR and MC1-4, depicted in Figure 3, we state that a link between GR and MC1-4 is not given. The ratings of both, male and female, differ for GR and MC1-4. However, in all subset groups, Tool C ranks 1st and Tool D ranks last which, as in the total group, allows us to distinct between best and imperfect tools. Therefore, Conjecture 2 must be rejected, but Conjecture 5 is supported for subset group 'Gender'.

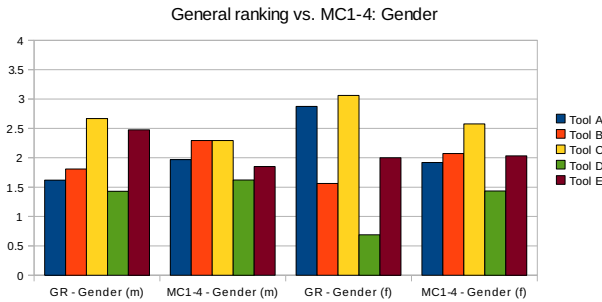


Fig. 3. Comparison of general ranking (GR) and mean of categories (MC1-4) for subset groups of 'Gender'

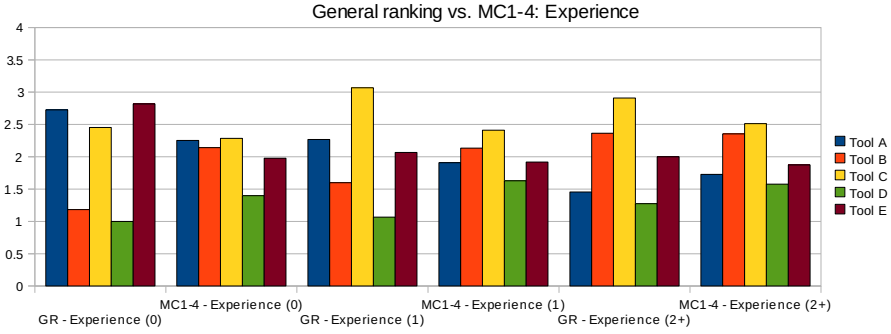


Fig. 4. Comparison of general ranking (GR) and mean of categories (MC1-4) for subset groups of 'Experience'

By Experience:

The subset groups for filter 'Experience' sum up to 6 groups, see Figure 4. For 'Experience (0)', the values and rankings of GR and MC1-4 differ, but considering users with at least little experience in group 'Experience (1)', we observe only a minor switch of ranks for Tool A and B. For users with high experience in group 'Experience (2+)', the values correspond and the ranking is identical.

Since the values of 'Experience (0)' do not correspond, we have to reject Conjecture 3. However, since the differences clearly diminish with higher experience of modeling users, Conjecture 6 is supported. Also, for both subset groups 'Experience (1)' and 'Experience (2+)', we can confirm that Conjecture 5 is supported since one can obviously inspect the best and the deficient tools chosen by the probands.

By Education:

The filter 'Education' also creates 6 subset groups. In Figure 5, the values are presented. We can state that for 'Education (0)', the values diverge, there are

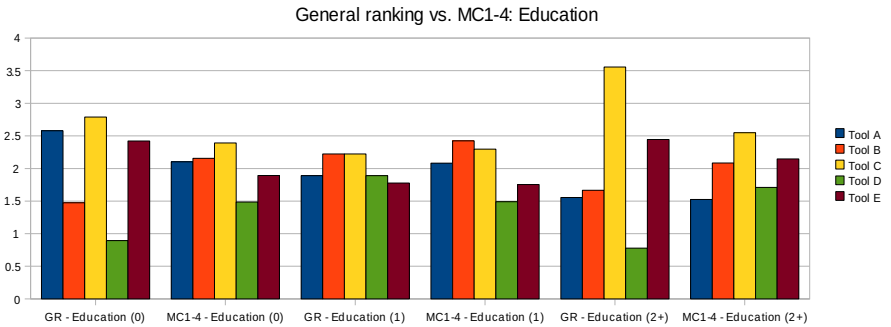


Fig. 5. Comparison of general ranking (GR) and mean of categories (MC1-4) for subset groups of education

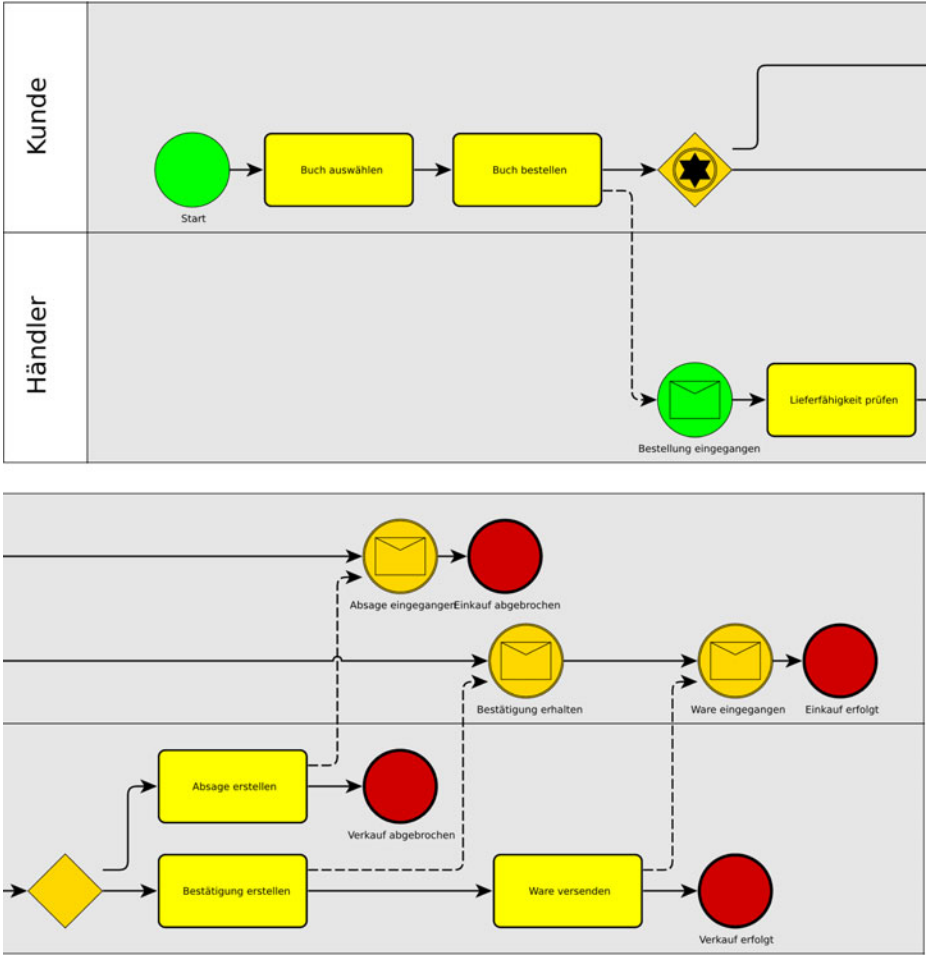


Fig. 6. Layout of Tool C scored highest in 7 of total 9 subset groups. For reading purposes, the image is for this paper manually cut in two halves of equal width; the upper first part is originally directly connected to the lower second part. As the study was conducted in German, the labels are given in German language.

two swaps of tools' ranking (Tool A,B and E). However, for groups with higher modeling education, we only notice one swap: Tool D and E for 'Education (1)', Tool A and D for 'Education (2+)'. Moreover, the distance between values that cause swaps diminish the higher the education level, e.g. the distance between Tool A and D for 'Education (2+)' is almost vanished to 0.19 on a 0-to-4-scale.

Also, as for subset group 'Experience', we cannot state that the corresponding Conjecture 4 for group 'Education' is supported. On the other hand, we can confirm an improvement of the results the higher the educational background, therefore Conjecture 7 is supported. Moreover, Conjecture 5 is supported for 'Education (0)' and clearly for 'Education (2+)', too.

Before closing the evaluation section, we sum up the results and conjectures of this section. Conjectures 1, 2, 3 and 4 are rejected since the differences between the values of the general ranking (GR) and the mean of categories (MC1-4) diverge in the total group and at least one subset group of the filters (gender, education, experience). However, we confirm Conjecture 5 for 7 of total 9 subset groups. The Conjecture states the correct prediction of the user preference tendency of a tool's layout capabilities when using our catalogue of statements. Also, Conjectures 6 and 7 can be confirmed. They state that the values of subset groups 'Experience' and 'Education' correspond for GR and MC1-4 when considering the groups with average or higher practice experience or at least basic education respectively for the field of business process modeling.

The layout from Tool C that was favoured by the probands in the general ranking is depicted in Figure 6.

5 Related Work

The term of secondary notation is due to fundamental cognitive research of [6,17]. Secondary notation is an important part of the cognitive dimensions of notation framework developed by [7]. It is also used in the study design of [5] where differences in understandability of business process models between novices and experts are targeted. In [18], graph aesthetics are used for cognitive measurements.

The area of research on layout aesthetics is broader if considering other diagram types or graph classes, too. There are diagram types widely related to BPMN, e.g. UML-diagrams which were part of an analysis in [19] where laws for diagram layout were formalized, e.g. the 'law of proximity'. This would refer to our statements 'ORTHOGONAL' and 'ELEMENT_SIZE'. Aesthetics for UML-diagrams were also proposed in the context of automatic layout in [20]. Suggestions of aesthetics are also given for Petri-Nets in [21] or for more general graphs in [22]. In [23], metrics are proposed and validated for entity-relationship (ER-) diagrams. Aesthetics, or 'effects', for social-network visualization was considered in [14] with the focus on edge crossings, in our case 'CROSSING'.

The set of layout aesthetics is further enriched by research works and studies of [8,9,24] where a subset of aesthetics is target of an ranking analysis. Further confirmation for aesthetics ranking is conducted by [25] that states that diagrams with short edges can be read more easily because the nodes' proximity is higher and the probability of crossings ('CROSSING') is lower. Also, [26] confirmed the importance of 'CROSSING'. Moreover, general modeling guidelines, e.g. [25], are available and provide a fundamental set of aesthetics.

User studies concerning aspects of layout aesthetics are done by [9] for UML, [15] for syntactical structures in process models and also [27] for general graphs. In [5], the focus is on secondary notation. The cognitive complexity of integrating multiple diagrams with different notations is examined in [28].

6 Conclusion

In this work, we analyzed secondary notation in terms of layout aesthetics and users' preferences of layout aesthetics for BPMN with consideration of inhomogeneous user groups. We proposed a catalogue of criteria that promise modeling results that are well-accepted by most users when being applied.

The formalized catalogue is tested by the conduction of a user study. The results of the study were presented and interpreted. The data analysis of the study results was performed with respect to not only all participants at a time but also to subset groups according to modeling experience, modeling education and gender. We were able to show that our layout catalogue is most appropriate when applied for users with average or higher practice experience and users with at least basic education in business process modeling. We also could show that our catalogue of statements is sufficient to predict the tendency of the users' judgement for the layout capabilities of a tool. For future works, we like to address a broader set of probands, not only in quantity but also in consideration of more diverse backgrounds, e.g. culture and/or nationality.

Concluding, our results can be used for designing powerful algorithms in modeling tools for BPMN that produce layout for BPMN diagrams that will be well-received by users.

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